## BI / read / 15

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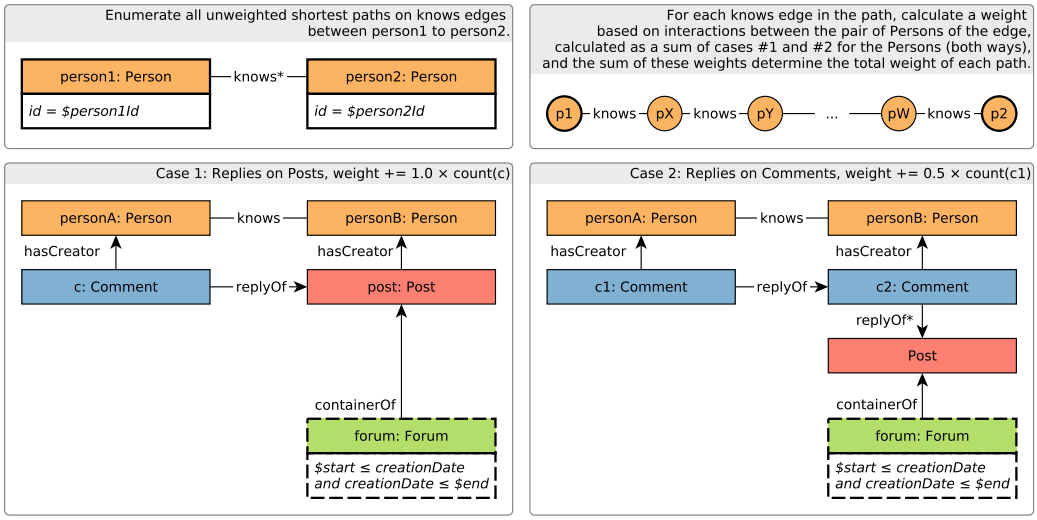
|---|---|
| query | BI / read / 15 |
| title | Trusted connection paths through forums created in a given timeframe |
| pattern |  |
| desc. | Given two Persons, find all (unweighted) shortest paths between these two Persons, in the subgraph induced by the knows relationship.<br>Then, for each path calculate a weight. The nodes in the path are Persons, and the weight of a path is the sum of weights between every pair of consecutive Person nodes in the path.<br>The weight for a pair of Persons is calculated based on their interactions:<br>• Every direct reply (by one of the Persons) to a Post (by the other Person) contributes 1.0.<br>• Every direct reply (by one of the Persons) to a Comment (by the other Person) contributes 0.5.<br>Only consider Messages that were created in a Forum that was created within the timeframe (interval) `[startDate, endDate]`. Note that for Comments, the containing Forum is that of the Post that the comment (transitively) replies to. Also note that interactions are counted both ways.<br>Return all paths with the Person IDs ordered by their weights descending. |

params

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|---|---|---|---|
| 1 | `person1Id` | ID | |
| 2 | `person2Id` | ID | |
| 3 | `startDate` | Date | |
| 4 | `endDate` | Date | |

result

| | | | | |
|---|---|---|---|---|
| 1 | `person.id` | [ID] | C | Ordered sequence of the Person IDs in the path |
| 2 | `weight` | 64-bit Float | C | |

sort

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|---|---|---|---|
| 1 | `weight` | ↓ | The order of paths with the same weight is unspecified |
| 2 | `personIds` | ↑ | The IDs in the paths are used for lexicographical sorting |

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| CPs | 1.2, 2.1, 2.2, 2.4, 3.3, 5.1, 5.3, 7.2, 7.3, 7.5, 7.7, 8.1, 8.2, 8.3, 8.4, 8.5, 8.6 |